Dylan Adams

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Education

Harvard University — Cambridge, MA
S.B. Candidate, Electrical Engineering; prospective secondary in Neuroscience — May 2030
Relevant Coursework: CS50, First Year Seminar "Letting Claude Cook"

Avon Grove High School — West Grove, PA
Class Rank 2/419; GPA 4.0 unweighted; SAT 1510 (Math 780); National Merit Finalist — June 2026
Relevant Coursework: AP Physics C, AP Computer Science A & Principles, AP Research, AP Statistics

Engineering Projects

CESyL – Custom Engineered Synthesizer and Lightbox — github.com/DylAdams/CESyL
Designer & Builder (hardware, firmware, software) — June 2023 – May 2025

- Built a standalone digital synthesizer pairing an Arduino microcontroller (Teensy 4.1) with a headless Raspberry Pi running Linux; 55 rotary encoders and ~300 RGB LEDs across 3 custom-made PCB designs
- Wrote C++ firmware that decodes Gray-code rotary encoder input and drives 900+ LED channels through chained shift registers with software PWM; defined the serial message protocol to the host
- Developed a multithreaded Python service that exchanges parameters with the Surge XT synth engine over OSC in both directions, with every control mapped from a JSON configuration; debugged OSC timing issues

Bobbert – Discord Music Bot — github.com/DylAdams/Bobbert
Developer (Python) — May 2022 – Present

- Built and maintain a Python Discord bot (discord.py, asyncio, yt-dlp) that streams audio with a playback queue

1970s Turntable Restoration — Independent Project
Repair & Restoration — 2023

- Restored a BSR turntable to daily use (designed/3d printed new arm-head compatibility piece; replaced rusted internal cables; installed new play head); shown on YouTube

Learning to Build with AI Coding Tools (Claude) — Self-Directed
Personal Projects — Sept. 2026 – Present

- Used Claude Code as a coding partner to prototype a real-time speech-to-face avatar (PyTorch, MediaPipe) and a hand-tracking chord controller for Ableton Live; practiced specifying behavior, testing, and measuring latency
- Learning Linux by setting up a Windows/Linux dual boot on a laptop, with Claude as a guide through BitLocker, Secure Boot, NVIDIA drivers, and disk repartitioning

Experience

Avon Grove School District — West Grove, PA
Summer Technology Intern — Summers 2023 – 2025

- Serviced and maintained ~5,000 computers across four school buildings with the district technology team

Leadership & Activities

Avon Grove High School Makerspace (Chief Science Officers Program) — West Grove, PA
Co-Founder, Student Planning Team — Dec. 2023 – June 2026

- Co-led 4 students with the superintendent and tech director to fund, design, and open a 2,000 sq ft makerspace; secured ~$10K in grant funding, ran a district-wide design contest, drafted equipment-certification policy

AGTV Broadcast Media Club & AGHS Theater — West Grove, PA
AGTV Co-Leader, Meteorologist & Producer; Theater Sound Tech Lead — Sept. 2022 – June 2026

- Led sound crews of up to 9 for Clue, The Wizard of Oz, and Mamma Mia!; co-led the 20-person broadcast club

Skills & Interests

Technical: C/C++, Python, Java; microcontrollers (Teensy/Arduino), Raspberry Pi/Linux; Git; circuit and PCB design (Fusion 360), soldering, prototyping
Interests: Songwriting and self-producing indie rock (Ableton Live), PC building